1. Evidence review: does the 2-isogeny-chain factor base help decomposition?

Decision DEC-20261015-a08389: **reject_scoped** H-ECDLP-60de36 · **evidence** EV-ECDLP-6bdff6 (contradicts, preliminary, toy tier) · **validation** VAL-20261015-1c3bff · **runs** RUN-ECDLP-f49f75-01, −02, −03 · **experiment** EXP-ECDLP-f49f75 · **goal** GOAL-ECDLP-001.

Scoped claim. For prime $p \equiv 3\ 4$ with $2^L \mid p+1$, $p \leq 23$ bits, and Semaev $m \in \{2,3\}$, encoding the factor-base set $V = x(R + \langle K \rangle)$ (the fibre of a chain of L Vélu 2-isogenies from $y^2 = x^3 + x$) by its chain equations makes Gröbner decomposition (Singular `std/slimgb` + `fglm`) **slower** than the dense encoding of the same set, and no faster than a random set of the same size; enumeration beats every Gröbner arm.

1.1. Question

The proposal (user, 2026-10-15) builds a P-256 factor base $\mathcal{B} = \{(u,v) : \Phi_x(u) = x(\Phi(R))\}$ from a 44-step 2-isogeny chain on $E_0 : y^2 = x^3 + x$ (P-256 has $2^{96} \mid p+1$), and needs a decomposition $T = P_1 + \ldots + P_6$ in $\tilde{O}(|\mathcal{B}|)$ for an $N^{\frac{1}{3}}$ attack. This is the Petit–Kosters–Messeng construction (PKC 2016, recalled). The experiment asks the cheapest question first: does the short chain description give a solver **any** advantage at toy scale?

1.2. Observation

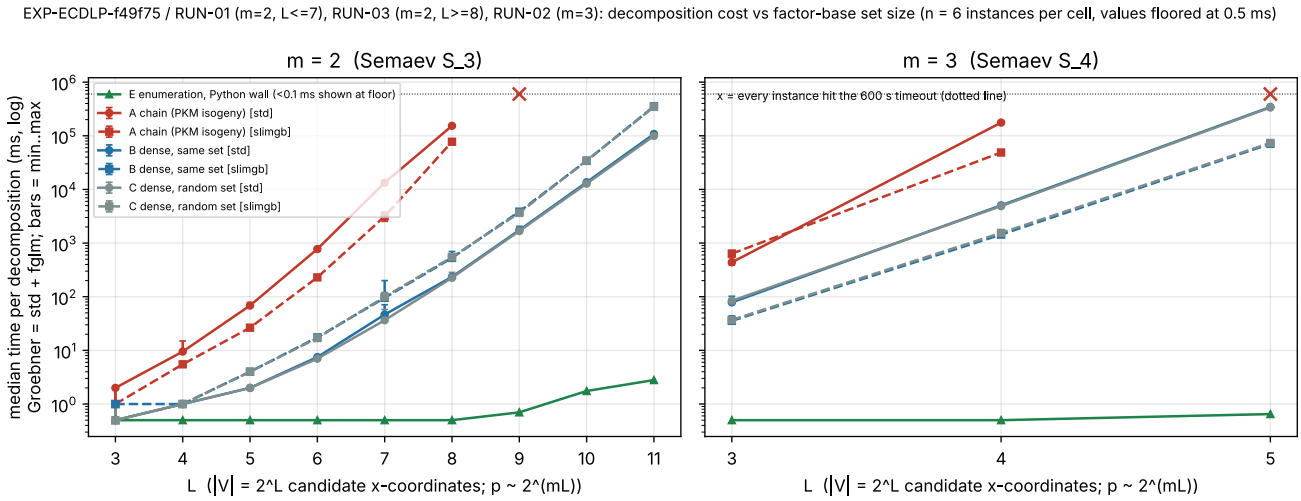


Figure 1: Median decomposition time vs L ($|V| = 2^L$), arms A (chain), B (dense, same set), C (dense, random set), E (enumeration); n = 6 per cell; × = all instances timed out at 600 s. Measured.

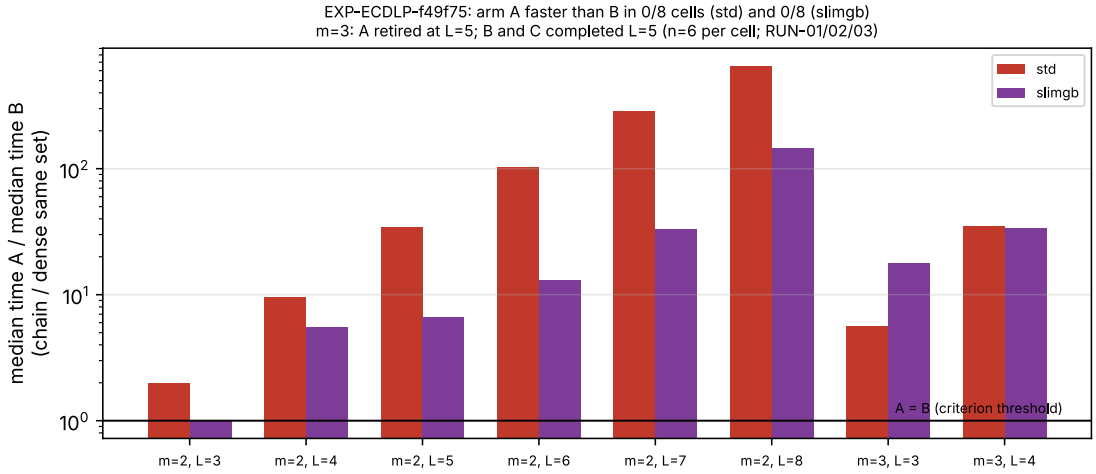


Figure 2: Frozen criterion 1: median time of A over B per jointly completed cell, both solvers (derived from measured medians; `criteria.json`). A threshold of 1 would be parity.

frozen criterion	**threshold**	**observed**
A faster than B, jointly completed cells	≥ 80% to support; A ≥ B in ≥ 80% falsifies	std: 0/8 faster; slimgb: 0/8 faster (7 slower, 1 tie at 1 ms)

m = 3 retirement	A completes larger L than B and C to support; A retired at L ≤ theirs falsifies	A retired at L = 5; B and C completed L = 5
positive control	100% planted in variety	248/248 (non-excluded cells)

1.3. Comparison

- The A/B gap grows with L: 2× → 654× (std) at m = 2, L = 3 → 8; 5.6× → 35× at m = 3, L = 3 → 4. The chain replaces one degree-2^L equation per point by L quadratics, multiplying the variable count by L; eliminating the auxiliary variables gives exactly arm B's polynomial (validator check), so A and B define the same variety.
- B and C are indistinguishable: the solver cannot see that V came from an isogeny.
- Enumeration (E) needs $2|\mathrm{FB}|^{m-1}$ group operations (629 at m = 3, L = 5) and is orders of magnitude cheaper than any Gröbner arm at every cell.

1.4. Inference

The chain encoding supplies no solver advantage at toy scale under this instrument; it is a growing slowdown. The isogeny structure lives in E_0's group law, which shares no homomorphism with the target curve's addition, and the experiment found no indirect effect either. The decomposition step the proposal needs remains unsolved (problem record `problems/pkm-isogeny-chain-factor-base-no-fast-decomposition.md`).

1.5. Limitations

- Toy scale ($p \leq 23$ bits), $m \leq 3$; P-256 needs $m = 6$, $L = 44$. One solver family (Singular); msolve/Magma F4 were not available.
- Process: RUN-01 Gröbner cells m = 2, L ≥ 8 invalidated by a launcher int overflow (AMD-02), re-run as RUN-03; m = 2, L = 12 and m = 3, L = 6 not executed (host 2 h limit; neither can affect the criteria). Manifests were written post-run; hand-typed amendment timestamps corrected in NOTE-01. Validator verdict `incomplete` on records only; its idle-host re-run of 48 instances matched every deterministic field and timings within 15%.
- 1 ms timer resolution at small L; four workers on four vCPUs.

1.6. Next actions

1. Keep the problem record open; next discriminating step: the m = 3 grid under an F4 solver with first-fall/ regularity degrees when msolve is available.
2. No further P-256 work on this route until a solver beats the dense and random controls on this frozen grid.